ME5847/MTE5847 – Homework 1: Due at 2pm on Oct 30th, 2026

1. A 0.1 cm^2 electrode with $C_d = 20\ \mu F/cm^2$ is subjected to a potential step under conditions where R_s is 1, 10, or 100 Ω. In each case, what is the time constant, and what is the time required for the double-layer charging to be 95% complete?

2. What are the half reaction and full reaction for the following cells?

a) Zn/Zn^{2+}, $Cl^-/AgCl/Ag$
b) $Pt/H_2/H^+$, $Cl^-/AgCl/Ag$
c) $Pt/Cl_2/Cl^-//Fe^{2+},Fe^{3+}/Pt$
d) $Ag/Ag^+//H^+/H_2/Pt$